LUND UNIVERSITY

Jinguo Sun

Division of Combustion Physics, Lund University, Lund, Sweden

E-mail: jinguo.sun@fysik.lu.se; TEL: +46-762190852

➢ Education

2022.09 – now **Post-doc researcher**	Division of Combustion Physics, Lund University (Advisor: Prof. Andreas Ehn)
2017.09 – 2022.07 **Ph. D**.	Department of Energy and Power Engineering, Tsinghua University **(**Supervisor: Prof. Shuiqing Li)
2019.12 – 2020.12 **Visiting PhD student**	Institute for Combustion Technology, RWTH Aachen (Advisor: Prof. Heinz Pitsch)
2013.09 – 2017.07 **B.E.**	Department of Energy and Power Engineering, Tsinghua University, Beijing, China

➢ Research Interests

- Plasma-assisted combustion for hydrocarbon and carbon-free fuels (NH_3).
- Laser-based optical diagnostics.
- Electric field/Plasma-flame interaction modeling.
- Combustion instability; Swirl flames dynamics.

➢ Publications

✓ 30+ papers in energy/combustion/plasma-related peer-reviewed journals and conferences, such as *Combustion and Flame*, *Proceedings of the Combustion Institute*, *Fuel*, *Energy & Fuels*, *Applications in Energy and Combustion Science*, *Renewable and Sustainable Energy Reviews*, *Langmuir*, etc.

✓ 4 patents (One U.S. patent and three China patents)

✓ Academic profiles

Google Scholar: https://scholar.google.com/citations?user=gsfMcpUAAAAJ&hl=en

Web of Science: https://webofscience.clarivate.cn/wos/author/record/GRS-8523-2022

ORCID: https://orcid.org/0000-0001-7932-4670

ResearchGate: https://www.researchgate.net/profile/Jinguo_Sun

- *Published*

1. **Jinguo Sun***, Sebastian Nilsson, Jonas Ravelid, Yupan Bao, Andreas Ehn. Fluorescence lifetime imaging of nitric oxide in nanosecond pulsed discharge-assisted NH_3/air flames. *Plasma Sources Science and Technology* 34.3 (2025): 035011.
2. Zhiyong Wu, Can Ruan, **Jinguo Sun**, Niklas Jüngst, Marcus Aldén, Zhongshan Li*. Visualization of unsteady combustion of single aluminum droplets: coalescence, eruption and fragmentation. *Combustion and Flame* 275 (2025): 114103.
3. **Jinguo Sun**, Yupan Bao*, Kailun Zhang, Alexander A. Konnov, Mattias Richter, Elias Kristensson, Andreas Ehn. A comprehensive study on dynamics of flames in a nanosecond pulsed discharge. Part II: Plasma-assisted ammonia and methane combustion. *Combustion and Flame* 275 (2025): 114076.
4. Yupan Bao*, Kailun Zhang, **Jinguo Sun**, Tomas Hurtig, Mattias Richter, Elias Kristensson, Andreas Ehn. A comprehensive study on dynamics of flames in a nanosecond pulsed discharge. Part I: Discharge formation and gas heating. *Combustion and Flame* 275 (2025): 114075.
5. Yipeng Li, **Jinguo Sun**, Qian Huang, Reinhold Kneer, Shuiqing Li*. Experimental investigation on the effects of air/fuel distribution on stability limit and NO_x emission of NH_3/CH_4 flame. *Combustion and Flame* 268 (2024): 113606.
6. Can Ruan, Zhiyong Wu*, **Jinguo Sun**, Niklas Jüngst, Edouard Berrocal, Marcus Aldén, Zhongshan Li. Ignition, stabilization and particle-particle collision in lifted aluminum particle cloud flames. *Proceedings of the*

Combustion Institute 40.1-4 (2024): 105596.

7. **Jinguo Sun***, Jonas Ravelid, Yupan Bao, Sebastian Nilsson, Alexander A. Konnov, Andreas Ehn. Dynamics of atomic oxygen production in an NH_3/air flame assisted by a nanosecond pulsed plasma discharge. *Proceedings of the Combustion Institute* 40.1-4 (2024): 105477.
8. **Jinguo Sun*,** Yupan Bao, Jonas Ravelid, Sebastian Nilsson, Alexander A. Konnov, Andreas Ehn. Plasma-assisted NH_3/air flame: Simultaneous LIF measurements of O and OH. *Combustion and Flame* 266 (2024): 113529.
9. Sebastian Nilsson, David Sanned, Adrian Roth, **Jinguo Sun**, Edouard Berrocal, Mattias Richter, Andreas Ehn*. Holistic analysis of a gliding arc discharge using 3D tomography and single-shot fluorescence lifetime imaging. *Communications Engineering* 3.1 (2024): 103.
10. Sam Taylor*, Filip Hallböök, Rob Temperton, **Jinguo Sun**, Lisa Rämisch, Sabrina Gericke, Andreas Ehn, Johan Zetterberg, Sara Blomberg*. In situ ambient pressure photoelectron spectroscopy study of the plasma-surface interaction on metal foils. *Langmuir* 40 (2024): 13950–13956.
11. **Jinguo Sun*,** Yupan Bao, Jonas Ravelid, Sebastian Nilsson, Alexander A. Konnov, Andreas Ehn. Application of emission spectroscopy in plasma-assisted NH_3/air combustion using nanosecond pulsed discharge. *Combustion and Flame* 263 (2024): 113400.
12. **Jinguo Sun**, Yihua Ren, Yong Tang, Shuiqing Li*. Influences of heat flux on extinction characteristics of steady/unsteady premixed stagnation flames. *Proceedings of the Combustion Institute* 38.2 (2022): 2305-14.
13. **Jinguo Sun**, Yong Tang, Shuiqing Li*. Plasma-assisted stabilization of premixed swirl flames by gliding arc discharges. *Proceedings of the Combustion Institute* 38.4 (2021): 6733-6741.
14. **Jinguo Sun**, Wei Cui, Yong Tang, Chengdong Kong, Shuiqing Li*. Inlet pulsation-induced extinction and plasma-assisted stabilization of premixed swirl flames. *Fuel* 328 (2022): 125372.
15. **Jinguo Sun**, Qian Huang, Yong Tang, Shuiqing Li*. Stabilization and emission characteristics of gliding arc-assisted NH_3/CH_4/air premixed flames in a swirl combustor. *Energy & Fuels* 36.15 (2022): 8520-8527.
16. **Jinguo Sun**, Hu Wu, Yong Tang, Chengdong Kong, Shuiqing Li*. Blowout dynamics and plasma-assisted stabilization of premixed swirl flames under fuel pulsations. *Applications in Energy and Combustion Science* 14 (2023): 100122.
17. Yupan Bao, Chengdong Kong, Jonas Ravelid, **Jinguo Sun**, Sebastian Nilsson, Elias Kristensson, Andreas Ehn*. Effect of a single nanosecond pulsed discharge on a flat methane-air flame. *Applications in Energy and Combustion Science* 16 (2023): 100198.
18. Yuanping Yang, Qian Huang, **Jinguo Sun**, Peng Ma, Shuiqing Li*. Reducing NO_x emission of swirl-stabilized ammonia/methane/air tubular flames through fuel-oxidizer mixing strategy. *Energy & Fuels* 36.4 (2022): 2277-2287.
19. Yong Tang, **Jinguo Sun**, Baolu Shi, Shuiqing Li, Qiang Yao*. Extension of flammability and stability limits of swirling premixed flames by AC powered gliding arc discharges. *Combustion and Flame* 231 (2021): 111483.
20. **Jinguo Sun,** Yupan Bao, Jonas Ravelid, Andreas Ehn*. Emission spectroscopy of nanosecond pulsed plasma discharges in ammonia/air flames. *25^{th} International Symposium on Plasma Chemistry*, Kyoto, Japan, 2023.
21. Jonas Ravelid, **Jinguo Sun**, Vassily Kornienko, Yupan Bao, Elias Kristensson, Andreas Ehn. Spatial and temporal investigation of atomic oxygen generation from ns-pulsed plasma using TAPLIF. *25^{th} International Symposium on Plasma Chemistry*, Kyoto, Japan, 2023.
22. Hu Wu, **Jinguo Sun**, Shaolong Li, Yong Tang, Wei Cui, Shuiqing Li*. Plasma-assisted stabilization of premixed swirl flames under flow pulsations, *13^{th} Asia-Pacific Conference on Combustion*, Abu Dhabi, United Arab Emirates, 2021.
23. **Jinguo Sun**, Wei Cui, Shuiqing Li*. Large-eddy simulation of premixed swirl flame dynamics under pulsating flow disturbances, *12^{th} Asia-Pacific Conference on Combustion*, Fukuoka, Japan, 2019.
24. **Jinguo Sun**, Qian Huang, Yuanping Yang, Yiyang Zhang, Shuiqing Li*. Stabilization and NOx emission of

gliding arc-assisted CH_4/NH_3 swirl flames. *Journal of Engineering Thermophysics* (in Chinese), 43 (2022): 2234-2241.

- ***Under peer-review***

25. Jonas Ravelid, **Jinguo Sun**, Vassily Kornienko, Alexander Konnov, Elias Kristensson, Yupan Bao, Andreas Ehn*. 2D imaging of atomic oxygen build-up and consumption after a nanosecond pulse discharge using light-field amplitude control. (Under review in *Plasma Sources Science and Technology*)
26. **Jinguo Sun***, Kailun Zhang, Yupan Bao, Christian Brackmann, Mattias Richter, Alexander A. Konnov, Andreas Ehn, Spatiotemporal Imaging of NH_2 in Plasma-Assisted NH_3 Combustion via Nanosecond Pulsed Discharge. (Under review in *Proceedings of the Combustion Institute*)

- ***Patents***

27. Shuiqing Li, Minhang Song, **Jinguo Sun**, Qian Huang. (2023): Atmosphere-adjustable multi-staged swirl ammonia burner. United States Patent (Patent No US011747014B2).
28. Shuiqing Li, Minhang Song, **Jinguo Sun**, Yuanping Yang, Qian Huang. (2021): Axial tangential multi-stage swirling ammonia-doped burner with adjustable atmosphere. China Patent (Patent No ZL202110923877.8)
29. **Jinguo Sun**, Liangliang Dai, Yuxin Ke, Xingfang Liang. (2020): An improved phosphorus diffusion furnace. China Patent (Patent No ZL201922245564.2)
30. Shuiqing Li, Hu Wu, Wei Cui, Yihua Ren, **Jinguo Sun**. (2019): Direct control method and equipment for swirl burner stability under plasma excitation. China Patent (Patent No ZL201910040680.2)

➢ Awards and Scholarships

- 2025, **Humboldt Research Fellowship**, Alexander von Humboldt Foundation.
- 2024, **Outstanding Paper Award**, Combustion Academic Annual Conference, Chinese Society of Engineering Thermophysics (Top 1% per year)
- 2022, two papers were selected as the **journal covers** of *Energy & Fuels* (Vol 36/Issues 4&15)
- 2021, **Comprehensive Scholarship (G1)** for Ph.D. students at Tsinghua University
- 2019, **Chinese Scholarship Council** for joint Ph.D. student
- 2014–2016 (three consecutive years), **Comprehensive Scholarship (G1)** for B.E. students at Tsinghua University

➢ International conferences

40th International Symposium on Combustion	2024.07	Milano, Italy
25th International Symposium on Plasma Chemistry	2023.05	Kyoto, Japan
13th Asia-Pacific Conference on Combustion	2021.12	Abu Dhabi, UAE
38th International Symposium on Combustion (**2 talks**)	2020.01	Adelaide, Australia
17th International Conference on Numerical Combustion	2018.05	Aachen, Germany
12th Asia-Pacific Conference on Combustion	2017.07	Fukuoka, Japan

➢ Peer-review activities

In the past three years, I have reviewed a total of **18** papers for *Combustion and Flame*, *Proceedings of the Combustion Institute*, *Renewable and Sustainable Energy Reviews*, *Fuel*, *Energy & Fuels*, *Combustion Science and Technology*, *International Journal of Hydrogen Energy*, *International Journal of Heat and Fluid Flow*, etc.

➢ Projects

LAPLAS: Advanced Laser Diagnostics for Discharge Plasma

2022.09–now *European Research Council (ERC)* *852394*

- Served as a main researcher.

Electrical and Plasma Assisted Combustion Instability Control for Jet Engine under Extreme Conditions

2017.01–2020.12. *National Natural Science Funds of China* *91641204*

- Served as a main researcher and wrote the completion report; also contributed to the proposal writing.

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➢ Fundings raised

Plasma-Assisted Ammonia Combustion Using a High Voltage Repetitive Pulse Generator

2024.11–2026.11 *The Märta and Eric Holmberg Endowment* ~25k Euro

- Principle Investigator.

Laser Diagnostics of Plasma-Assisted Ammonia Combustion

2023.11–2025.11 *The Märta and Eric Holmberg Endowment* ~20k Euro

- Principle Investigator.

➢ Skills

- **Languages**: Chinese Mandarin (Native), English (Proficient)
- **Experiments:** Various optical diagnostics, including LIF, Scattering, PIV, TDLAS, and chemiluminescence, *etc.*
- **Simulations**: OpenFOAM, CHEMKIN, Cantera & ZDPlasKin (Proficient), COMSOL & Fluent (Intermediate).
- **Programming**: Matlab & Python (Proficient), C++ & Fortran (Intermediate).
- **Data processing**: Proper orthogonal decomposition, Abel-inversion, 3D reconstruction, *et al.*